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Next item →

1.

In the continuous-time state-space model form:
 $\dot{x}(t) = Ax(t) + Bu(t) + w(t)$
 $z(t) = Cx(t) + Du(t) + v(t)$
what are the names of the matrices A , B , C , and D ?

☐ A is the "dynamics matrix," B is the "process matrix," C is the "sensor matrix," and D is the "feedforward matrix."

☐ A is the "evolution matrix," B is the "input matrix," C is the "combination matrix," and D is the "instantaneous matrix."

☐ A is the "system matrix," B is the "feedthrough matrix," C is the "output matrix," and D is the "input matrix."

☒ A is the "system matrix," B is the "input matrix," C is the "output matrix," and D is the "feedthrough matrix."

Correct

Yes, this is correct.
2.

What kind of state-space model should I use if I wish to track a target that tends to move in nearly circular paths?

☒ A "coordinated turn" (CT) model.

☐ A "nearly constant position" (NCP) model.

☐ A "nearly constant velocity" (NCV) model.

☐ A "nearly constant acceleration" (NCA) model.

Correct

Yes, this is correct.
3.

Run the Coursera Labs Jupyter notebook for Lesson 1.2.2 titled "Jupyter notebook used in conjunction with practice quiz." This notebook implements simulations of different models commonly used in target-tracking applications.

Do not modify the code line "randn('state',0)". This operation forces the pseudo-random number generator in Octave to generate a reproducible sequence of random values, which is needed for the notebook to generate answers that can be verified by the quiz grader.

For this question, simply execute the lab and enter the final y -coordinate value (the y variable) for the NCP simulation. Enter three digits after the decimal point.

-0.033038

Correct

Yes, this is correct.
4.

What is the correct syntax for implementing e^{At} in Octave if A is a matrix, t is a scalar, and x_0 is a vector?

☐ `exp (A*t)`

☒ `expm (A*t)`

☐ `exp (A*t) *x0`

☐ `expm (A*t) *x0`

Correct

Yes, this is correct.
5.

Suppose that A is 2×2 in size. Which figure(s) below could be the response e^{At} ? (Select all that apply.)

Figure (A)

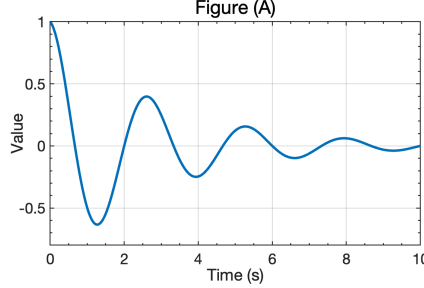


Figure (B)

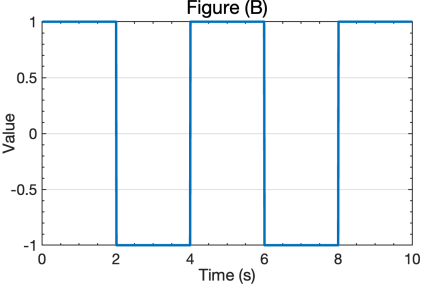


Figure (C)

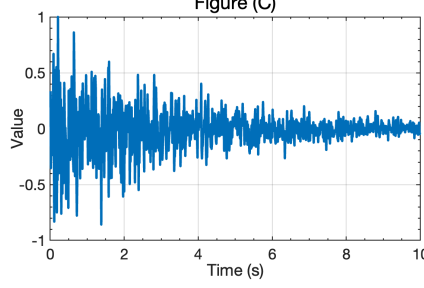
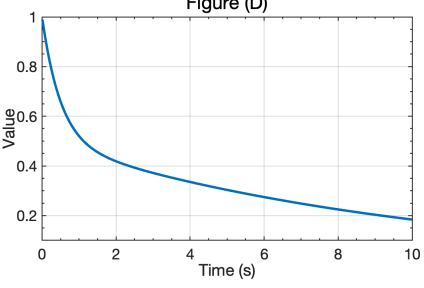


Figure (D)



☒ Figure (A).

Correct

Yes, this could be the response to e^{At} for some value of A .

☐ Figure (B).

☐ Figure (C).

☒ Figure (D).

Correct

Yes, this could be the response to e^{At} for some value of A .

6.

Suppose that A has eigenvectors:
 $v_1 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$.

What transformation matrix T will diagonalize the A matrix via the computation $\Lambda = T^{-1}AT$?

☐ $T = \begin{bmatrix} 0.2 & 0.6 \\ 0.4 & 0.8 \end{bmatrix}^{-1}$.

☐ $T = \begin{bmatrix} 0.2 & 0.4 \\ 0.6 & 0.8 \end{bmatrix}$.

☐ $T = \begin{bmatrix} 0.2 & 0.4 \\ 0.6 & 0.8 \end{bmatrix}^{-1}$.

☒ $T = \begin{bmatrix} 0.2 & 0.6 \\ 0.4 & 0.8 \end{bmatrix}$.

Correct

Yes, this is correct.

7.

If A , B , C , and D are the state-space matrices of a continuous-time model, and A_d , B_d , C_d , and D_d are the corresponding discrete-time state-space matrices, which of the following statements are true in general? (Select all that apply.)

☐ $A_d = A$.

☐ $B_d = B$.

☒ $C_d = C$.

Correct

Yes, this is correct.

☒ $D_d = D$.

Correct

Yes, this is correct.

8.

If A is the system matrix of a continuous-time model, how can we compute the corresponding A_d of a discrete-time model of the same system? Assume that the sampling period is Δt and select all that apply.

☐ $A_d = A$.

☐ $A_d = \exp(A * \Delta t)$.

☒ $A_d = \expm(A * \Delta t)$.

Correct

Yes, this is correct.

☐ $A_d = I + A\Delta t$.

9.

When comparing the simulation of a continuous-time model to a simulation of the corresponding discrete-time model (for the same inputs), which of the following is true?

☐ The states and outputs of both models are identical at all points in time t .

☐ The states of both models are identical at the sampling points $t = k\Delta t$ and the outputs are identical at all points in time t .

☐ The outputs of both models are identical at the sampling points $t = k\Delta t$ and the states are identical at all points in time t .

☒ The states and outputs of both models are identical at the sampling points $t = k\Delta t$ but are not identical at other points in general.

Correct

Yes, this is true.

10.

A discrete-time system is described by the (discrete-time) matrices:
 $A = \begin{bmatrix} 0.2 & 0.4 \\ 0.6 & 0.8 \end{bmatrix}$, $B = \begin{bmatrix} 1.0 \\ 1.2 \end{bmatrix}$, $C = \begin{bmatrix} 1.4 & 1.6 \end{bmatrix}$, and $D = 1.9$.

What is its observability matrix?

☒ $\mathcal{O} = \begin{bmatrix} 1.4 & 1.6 \\ 1.24 & 1.84 \end{bmatrix}$.

☐ $\mathcal{O} = \begin{bmatrix} 1.4 & 1.24 \\ 1.6 & 1.84 \end{bmatrix}$.

☐ $\mathcal{O} = \begin{bmatrix} 1.0 & 0.68 \\ 1.2 & 1.56 \end{bmatrix}$.

☐ $\mathcal{O} = \begin{bmatrix} 1.0 & 1.2 \\ 0.68 & 1.56 \end{bmatrix}$.

Correct

Yes, this is correct.